GOOD AND BAD INTERFACE: BMW IDRIVE VS LEXUS REMOTE TOUCH

Assignment 1







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BMW iDrive vs Lexus Remote touch

Bad UI example: Lexus Remote Touchpad (2014–2021)

The Lexus remote touchpad, the laptop-style trackpad used to control the central screen in most Lexus models from 2014 to 2021, is a clear case of bad interface design, because it made simple tasks slow, error-prone and most-importantly, distracting while driving.

A note on sources

I want to clarify that I have not used this system myself. This analysis is based on published reviews by automotive journalists and my personal interest in cars, through which I was always reminded of how difficult the Lexus system was to use. The criticism was so consistent that the touchpad became a well-known example of poor in-car interface design.

Objective of the interface

The Remote Touchpad lets the driver control the car's central infortainment system, which includes navigation, audio, phone, vehicle settings, etc. In most models, especially the first iterations, the screen was not touch-sensitive, so nearly every function went through the touchpad (Cars.com, 2016).

To explain, it is a small multi-touch pad on the centre console, used like a laptop trackpad. Sliding a finger moves a cursor on screen, pressing the pad selects, and pinching zooms the map. A few hardware buttons (such as home and menu) sit beside it, as you can see on the image below.



Figure 1: Lexus touchpad used from 2014 to 2020 (Lexus UK Media site, 2024)

The system was used in many vehicles. It debuted around 2014 on the RC and NX (Cars.com, 2015) and spread across the range, including the UX, LS, LC and, from 2020, the RX (Consumer Guide, 2019).

One article I found particularly interesting is Brian Normile's review of the Lexus LC 500 for Cars.com (Lexus LC500, 2019). He presents the LC 500 as one of the last great cars of its era: although it sold in small numbers, it marks the end of the ICE-period for Lexus as the car industry shifts towards electric vehicles. Even so, he singles out the remote touchpad as a weak point, saying it has long been one of his biggest deal-breakers.
In his view, the car is good enough to make him tolerate the touchpad, which shows how much the interface weighs on the experience of an otherwise excellent car.

The specific bad aspect

The problem with this particular interface is using a relative touchpad to steer a cursor onto small on-screen targets while driving. The menus and graphics had their own issues, but this input method is what made the system so hard to use.

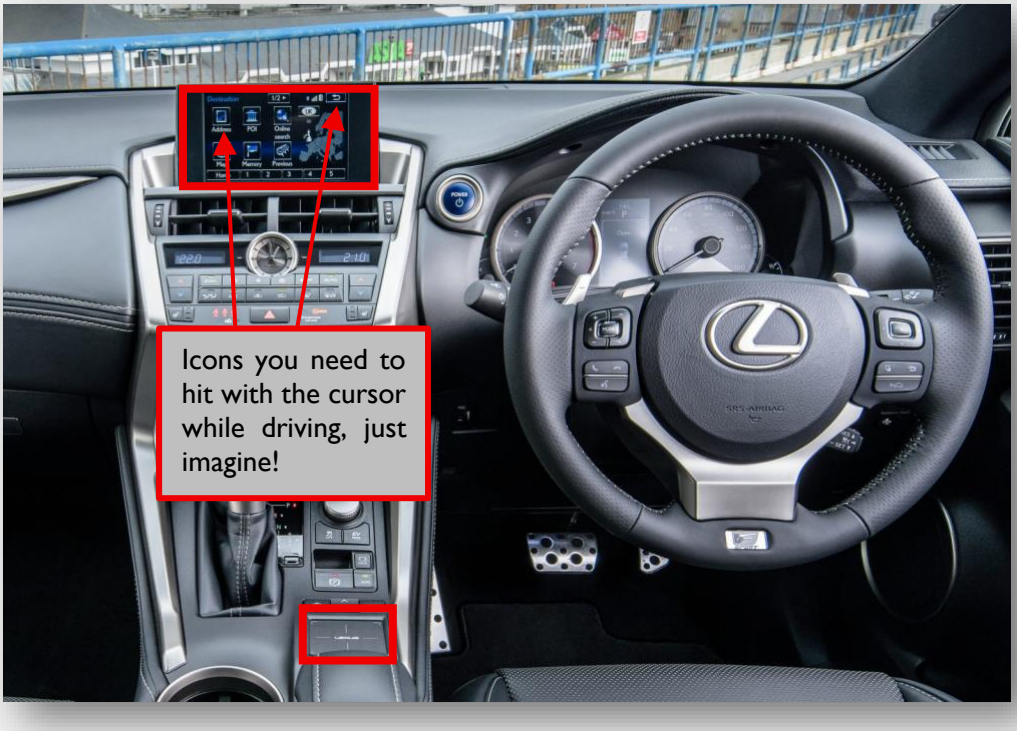


Figure 2: Interior description of Lexus infotainment (Lexus UK Media site, 2014)

Why it is bad

This system is bad because it requires the drivers' eyes and their full attention. A touchpad is flat and featureless, so the finger gets no clue where it is. The only feedback is the cursor on screen. Every selection pulls the driver's eyes off the road until the cursor reaches its target. Owners and journalists described it as impossible to use accurately while driving.

Mat Watson from Carow described it quite humoristic in his review of the Lexus LC500: '…when you encounter the horrible infotainment system, which you need a Masters degree in computer

science and the hand-eye coordination of a professional ping pong player just to use.' (Carwow, no date)

The desgin works against Fitt's law, as it requires precise aiming in a moving car. Selection gets harder as targets get smaller, and a fingertip on a small pad is easily thrown off by bumps and vibration.

Cars.com editors found the pad sometimes too sensitive and sometimes not sensitive enough, with false clicks opening menus they did not want (Cars.com, 2016). Each accidental selection then has to be undone, which costs more attention. Additionally, reviewers criticised the large number of features in the menus (Hick's law) that made the touchpad even more painful to use (Cars.com, 2016). More steps per task means more time looking away from the road.

A laptop trackpad works because the user is still and looking at the screen. A reviewer noted the Lexus pad felt like it came from a laptop and was unfriendly at highway speed (Digital Trends, 2020). Lexus tried to integrated desktop interaction, but in the wrong conecpt.

The result showed in the data: in a 2019 Consumer Reports owner survey, the Remote Touchpad had the lowest satisfaction of the built-in systems rated, at 46% (Cars.com, 2019).

Why it was probably designed this way

The individual decisions made sense on their own, the problem is how Lexus combined it in a driving context.
First, the screen was mounted high and far back, close to the driver's line of sight but out of reach, so it needed a remote controller. When Lexus later added touch to the RX, it had to redesign the dashboard so people could reach the screen.

Second, Lexus promoted the pad as working just like a laptop trackpad, and used the world's familiarity with phones and laptops as an argument for the remote touch interface. It included taps and pinch-to-zoom, and was described as the largest (and probably the only…) of its kind in the industry. The flaw was assuming a desk-based interaction transfers to a moving car, where attention belongs on the road.

Third, Japanese drivers were reportedly attached to the system, and a Lexus US team suspected internal survey answers were inflated by politeness (The Drive, 2021). This shows why usability testing needs real tasks and honest, independent users without bias.

Last, cars are planned years ahead, while consumer technology changes fast, which left Lexus stuck with an interface customers had already moved past (WardsAuto, 2021). Car interfaces are always behind in technology compared to our phones, laptops, etc.

Suggested improvements

1) A touchscreen within reach for tasks like entering an address or zooming a map, where direct pointing is fastest and needs no cursor.
2) Physical controls for frequent functions! Volume, temperature, climate control and media should have dedicated buttons or knobs that work by feel, without looking. This is a recurring problem for the entire car market right now, where every brand moves to button-less interiors with huge screens.
3) Tactile structure on the pad, such as ridges, edges or fixed zones (for example, swipe up/down to scroll a list, never free aiming) would let the finger find its place without the eyes.
4) Bigger targets and simpler menus, so fewer, larger steps are needed per task.

5) Stable, fast response by removing the lag and reducing sensitivity while the car is moving to prevent false clicks.
6) Phone mirroring by including Apple CarPlay and Android Auto as alternatives.

Lexus moved in this direction itself. The 2020 RX added a touchscreen moved closer to the driver, which reviewers said made the touchpad almost unnecessary (J.D. Power, no date). From the 2022 NX, Lexus dropped the touchpad entirely for a new touchscreen-based Lexus Interface (WardsAuto, 2021).



Figure 3: Lexus interior nowadays (Topspeed, 2023)

Good UI example: BMW iDrive

BMW iDrive, in its versions from the mid-2010s onward, is a good example of a good interface including redundant input. The driver can do almost any task with the touchscreen, the rotary controller, the touchpad on top of the controller, hand gestures, or voice, and choose whichever suits the moment.

A note on sources and version

This analysis is based on BMW's own documentation, automotive reviews and my long-term personal use. I have personally used all new versions of this interface since 2019. It covers the iDrive generations that added a touchscreen (from 2015). The original 2001 iDrive is a different story as it was widely considered a usability failure, which makes the later redesign an interesting case of a company learning from its mistakes.

Objective of the interface

iDrive is BMW's infotainment system, which controls navigation, media, phone, vehicle settings, etc. from one central screen (and head-up display and driver's screen in newer models). It debuted in the E65 7 Series in 2001, replacing a dashboard full of buttons with a single push-and-turn dial on the centre console (BMW Group, no date).

It evolved in stages:

Year	Change
2001	Rotary controller and central display; voice control for selected functions (BMW Group, no date)
Later 2000s	Direct buttons added around the controller for the most-used menus
2012	Touchpad on top of the controller, for handwriting letters and moving maps (BMWBlog, 2012)
2015	Touchscreen added in the 7 Series, alongside the controller.
2018 onwards	Touchscreen, hand gestures, voice, digital driver's display, head-up display

Many people thought iDrive was designed by Apple, including me a while ago. But no, iDrive was developed by BMW. The idea that Apple was involved comes from rumours around 2007 that BMW would replace iDrive with an Apple-designed interface (Gizmodo, 2007). The 'i' in the name probably adds to the confusion. Apple's only link today is CarPlay, which runs wirelessly inside iDrive as one more option.

The specific good aspect

Redundancy. When BMW added the touchscreen, it did not remove anything. BMW described the touch input as fully redundant. Everything can be done either with the controller or by touching the screen (CarAdvice, 2015).

The driver can use:

- The touchscreen, for direct pointing, such as tapping a song, typing an address, pinching to zoom a map.
- The rotary controller, turned to move between items and pressed to select, with shortcut buttons around it.
- The touchpad on the controller, for writing letters with a finger or moving the map.
- Voice commands, for hands-free tasks such as setting a destination.

- Hand gestures, for adjusting the volume, for example, pointing one finger and moving the hand in a circle.
- Two different screens at one. The driver can divide the screens by two applications, having the navigation on the left side, and music on the right side, for example.

The user can even switch method in the middle of a task, for example while entering an address.

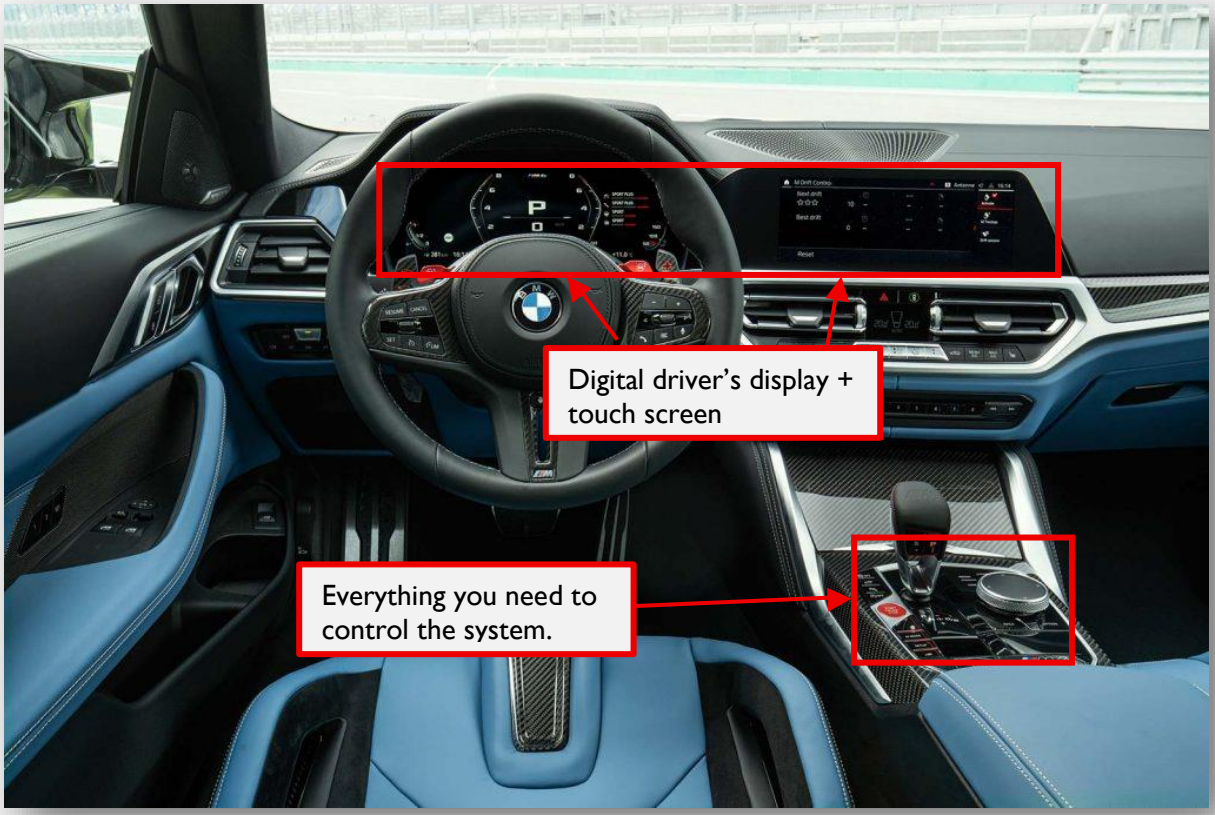


Figure 4: BMW interior from 2020 (MegaAutos, 2020)

Why it is good

This infotainment system system is good because each method fits a different context. Touch is fastest and most natural when the car is stopped. BMW found customers loved using it then (CarAdvice, 2015). While moving, the rotary controller works by feel and muscle memory, keeping eyes on the road (BMWBlog, 2017). Voice frees both hands and eyes. No single input is ideal everywhere, so offerin, 2015g several lets users pick the right one. Additionally, the controller suits driving (Fitts's law and feedback). Turning the knob moves exactly one item per click, so there is no small target to aim at. The physical clicks give tactile feedback, so the driver only needs a quick glance to confirm.



Figure 5: BMW X5 iDrive (AutoEvolution, 2022)

The system is flexible and efficienct to use (follows Nielsen's usability principles). Beginners can tap the screen like a phone, wheras experienced users can operate the controller and shortcut buttons without looking. In addition, the system is consistent The same menus respond to every input method, so learning the system once works for all of them. If one method does not work well for a person or situation (gloves, a bumpy road, a noisy cabin for voice), another is available.

Comparison with Lexus Remote Touch

Both systems faced the same problem: a screen the driver must operate while driving. Lexus offered one method, a free cursor operated by a slow-ish toucpad, needing the eyes and precise aim. BMW kept a controller that works by feel and added touch, hand gestures and voice on top. The contrast shows that the choice of input device matters as much as the screen design.

Limitations

No interface is perfect, and acknowledging this makes the analysis stronger, because the original iDrive was a failure. The 2001 version was widely criticised as slow and over-complicated (BMWBlog, 2017). One trade publication even compared its launch to a major corporate disaster (Electronic Design, 2004).



Figure 6: 2005 BMW iDrive (Wikipedia Commons, 2005)

The good design described here (after 2015) took over a decade of iteration.
To clarify, the voice method is the weakest method. Drivers were frustrated when it misunderstood commands, which is why it works as a complement, surely not a replacement.
To conclude, altough the BMW infotainment of the 21st is leagues above the Lexus's, redundancy has costs. More input hardware means more cost, more parts can fail and also more for users to discover, and menus must be designed to work well with all methods at once.

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